

Simplicial Complex Multiset Embedders

Algorithms 1 and 2: How They Work and How They Differ

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1 The Problem

We are given a *multiset* M of n vectors in $\mathbb{R}^k$. We represent M as a $k \times n$ matrix whose *columns* are the vectors; column order is arbitrary. We want a function

$$f : \mathbb{R}^{n \times k} \longrightarrow \mathbb{R}^N$$

that is:

- **Multiset-invariant**: permuting the columns of M does not change $f(M)$.
- **Injective (embedding)**: distinct multisets map to distinct outputs, i.e. f is an *embedder*, not merely an encoder.
- **Continuous (C^0)**: f is continuous in the entries of X .
- **Positively homogeneous of degree 1 (HomDeg1)**: $f(\lambda X) = \lambda f(X)$ for all $\lambda > 0$.

The two algorithms described here achieve this with output sizes:

	Algorithm 1	Algorithm 2
Output size N	$2nk + 1$	$2nk + 1 - k$

Algorithm 2 saves exactly k dimensions. Both are C^0 and HomDeg1 but *not* C^1 (piecewise-linear kinks appear where sorted order changes).

2 Shared Building Blocks

Both algorithms use the same core data structures and sub-steps.

Labels: Eji

Each of the nk matrix entries is identified by a label $e_{j,i}$, where $j \in \{0, \dots, n-1\}$ is the *column* (vector) index and $i \in \{0, \dots, k-1\}$ is the *row* (coordinate) index. The set of all labels is $E = \{e_{j,i} : 0 \leq j < n, 0 \leq i < k\}$.

Vertices: Maximal Simplex Vertex (MSV)

An MSV is a subset $V \subseteq E$; equivalently, a $\{0, 1\}$ -valued $n \times k$ matrix whose 1-entries mark the members of V .

Linear Combinations: Eji_LinComb

An *Eji_LinComb* L is an $n \times k$ matrix of non-negative integer counts obtained by adding MSVs:

$$L = V_{(0)} + V_{(1)} + \dots + V_{(\ell-1)}, \quad L[j][i] = \#\{s : e_{j,i} \in V_{(s)}\}.$$

A companion integer ℓ (the *index*) records how many MSVs were summed.

Canonical Form

Given an Eji_LinComb L , its *canonical form* $\tilde{L}$ is obtained by sorting the columns of L 's count matrix lexicographically (one column per vector, consistent with the column-vector convention). This mods out the arbitrary labelling of vectors (the j -indices), achieving multiset invariance: two inputs that are column-permutations of each other produce the same canonical Eji_LinCombs.

Hash to Hypercube

Each canonical Eji_LinComb $\tilde{L}$ is mapped deterministically to a pseudorandom point

$$h(\tilde{L}) \in [0, 1]^N$$

by hashing its count matrix and index with MD5 and converting the digest bits to floating-point coordinates. Distinct canonical Eji_LinCombs are (with probability 1 over the choice of hash function) mapped to distinct, “generically positioned” points.

Barycentric Subdivision

This sub-step, shared identically by both algorithms, converts a sorted list of gap-MSV pairs into a list of coefficient-Eji_LinComb pairs.

Input: m pairs $(\delta_0, V_0), \dots, (\delta_{m-1}, V_{m-1})$ with $\delta_0 \geq \delta_1 \geq \dots \geq \delta_{m-1} \geq 0$.

Output: m pairs (α_i, L_i) where

$$\alpha_i = (i+1)(\delta_i - \delta_{i+1}), \quad \delta_m := 0,$$

$$L_i = V_0 + V_1 + \dots + V_i \quad (\text{Eji_LinComb with index } i+1).$$

Key properties:

- $\alpha_i \geq 0$ (since gaps are sorted descending).
- $\sum_i \alpha_i = \sum_i \delta_i$ (by Abel summation; the total weight is preserved).
- The L_i are monotone: $L_{i+1} = L_i + V_{i+1} \geq L_i$ entrywise.

Final Output Assembly

Given the barycentric pairs (α_i, L_i) , each L_i is canonicalised and hashed to a point $p_i = h(\tilde{L}_i) \in [0, 1]^{N_{\text{hash}}}$. The main body of the embedding is then

$$\sum_{i=0}^{m-1} \alpha_i p_i \in \mathbb{R}^{N_{\text{hash}}},$$

to which a small number of *anchor values* (global or per-column extrema) are appended or prepended to fix the absolute scale.

3 Algorithm 1: Global Sort

Source file: `C0HomDeg1_simplicialComplex_embedder_1_for_array_of_reals_as_multiset.py`

Output size: $N = 2nk + 1$

Steps

1. **Global sort.** Treat all nk matrix entries as labelled scalars and sort them in *descending* order:

$$v_0 \geq v_1 \geq \dots \geq v_{nk-1}, \quad v_r = X[j_r][i_r], \quad e_r = e_{j_r, i_r}.$$

2. **Gaps and prefix MSVs.** Compute the $nk - 1$ consecutive differences $\delta_r = v_r - v_{r+1}$ for $r = 0, \dots, nk - 2$. Associate each gap r with the *prefix MSV* $V_r = \{e_0, e_1, \dots, e_r\}$ (the set of labels of the top $r+1$ values).
3. **Re-sort by gap size.** Sort the $nk - 1$ pairs (δ_r, V_r) in descending order of δ_r .
4. **Barycentric subdivision** (Section 2) with $m = nk - 1$.
5. **Canonicalise and hash.** For each (α_i, L_i) , compute $\tilde{L}_i$ and $p_i = h(\tilde{L}_i)$ with $N_{\text{hash}} = 2(nk - 1) + 1 = 2nk - 1$.
6. **Assemble output.**

$$f_1(X) = \left[\underbrace{\sum_i \alpha_i p_i}_{2nk-1 \text{ reals}}, \underbrace{v_0}_{\text{global max}}, \underbrace{v_{nk-1}}_{\text{global min}} \right] \in \mathbb{R}^{2nk+1}.$$

Note on the maximum. Given the global minimum and all $nk - 1$ gaps, one can reconstruct all nk values; storing the global maximum is technically redundant. It is included for implementation simplicity (a TODO comment in the code acknowledges this for degenerate cases).

Worked Example: $n = 4, k = 2$

Input multiset (four 2-vectors shown as column vectors):

$$M = \left\{ \begin{pmatrix} 4 \\ 2 \end{pmatrix}, \begin{pmatrix} -3 \\ 5 \end{pmatrix}, \begin{pmatrix} 8 \\ 9 \end{pmatrix}, \begin{pmatrix} 2 \\ 7 \end{pmatrix} \right\}$$

Step 1–2: Global sort and gaps.

Rank r	Value v_r	Label e_r	Gap $\delta_r = v_r - v_{r+1}$
0	9	$e_{2,1}$	1
1	8	$e_{2,0}$	1
2	7	$e_{3,1}$	2
3	5	$e_{1,1}$	1
4	4	$e_{0,0}$	2
5	2	$e_{0,1}$	0
6	2	$e_{3,0}$	5
7	-3	$e_{1,0}$	—

Global max = 9, global min = -3. The 7 prefix MSVs are $V_r = \{e_0, \dots, e_r\}$; for example $V_2 = \{e_{2,1}, e_{2,0}, e_{3,1}\}$.

Step 3: Re-sort by gap size. After sorting the 7 pairs by δ descending: gaps of size 5, 2, 2, 1, 1, 1, 0.

Steps 4–5: Barycentric subdivision. Non-zero α_i values (zero terms contribute nothing to the sum):

$$\alpha_0 = 1 \cdot (5 - 2) = 3, \quad \alpha_2 = 3 \cdot (2 - 1) = 3, \quad \alpha_5 = 6 \cdot (1 - 0) = 6.$$

The remaining $\alpha_i = 0$.

Step 6: Output. $N_{\text{hash}} = 2(8 - 1) + 1 = 15$.

$$f_1(X) = 3p_0 + 3p_2 + 6p_5 + [9, -3] \in \mathbb{R}^{17}.$$

where each $p_i \in [0, 1]^{15}$ is the hash of the corresponding canonical Eji_LinComb. The full numerical output is:

$$[7.07, 6.91, 2.78, 3.50, 1.01, 5.51, 4.07, 4.90, 10.02, 9.51, 5.13, 2.97, 8.04, 5.34, 7.41, 9, -3].$$

4 Algorithm 2: Per-Component Sort

Source file: COHomDeg1_simplicialComplex_embedder_2_for_array_of_reals_as_multiset.py

Output size: $N = 2nk + 1 - k$

Steps

1. **Per-column sort.** For each component $i = 0, \dots, k-1$, sort the n values in column i in *descending* order:

$$X[\sigma_i(0)][i] \geq X[\sigma_i(1)][i] \geq \dots \geq X[\sigma_i(n-1)][i].$$

2. **Per-column gaps and prefix MSVs.** For each column i , compute the $n - 1$ consecutive differences within that column:

$$\delta_r^{(i)} = X[\sigma_i(r)][i] - X[\sigma_i(r+1)][i], \quad r = 0, \dots, n-2.$$

Associate gap r in column i with the prefix MSV $V_r^{(i)} = \{e_{\sigma_i(0),i}, \dots, e_{\sigma_i(r),i}\}$ (all entries *within* column i ; no cross-column mixing). Record the k column minima: $m_i = X[\sigma_i(n-1)][i]$.

3. **Flatten and re-sort.** Collect all $k(n-1) = nk - k$ pairs $(\delta_r^{(i)}, V_r^{(i)})$ into one list and sort descending by gap size.
4. **Barycentric subdivision** (Section 2) with $m = nk - k$.
5. **Canonicalise and hash.** For each (α_i, L_i) , compute $\tilde{L}_i$ and $p_i = h(\tilde{L}_i)$ with $N_{\text{hash}} = 2k(n-1) + 1 = 2nk - 2k + 1$.
6. **Assemble output.**

$$f_2(X) = \left[\underbrace{m_0, \dots, m_{k-1}}_{k \text{ col. minima}}, \underbrace{\sum_i \alpha_i p_i}_{2nk-2k+1 \text{ reals}} \right] \in \mathbb{R}^{2nk+1-k}.$$

Why no column maxima? The maximum of each column is implicitly encoded: it equals the column minimum plus the sum of that column's gaps (which are embedded in the hash-weighted sum). Only the minima are needed as anchors to fix the absolute scale.

Worked Example: $n = 4, k = 3$

Input multiset (four 3-vectors shown as column vectors):

$$M = \left\{ \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} -3 \\ 5 \\ 1 \end{pmatrix}, \begin{pmatrix} 8 \\ 9 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 7 \\ 2 \end{pmatrix} \right\}$$

Row-0 (coordinate 0) minima: $m_0 = -3$, row-1: $m_1 = 2$, row-2: $m_2 = 1$.

Steps 1–2: Per-column sorts and gaps.

Col. i	Sorted desc. (value, label)	Gaps δ	Prefix MSVs (for each gap)
0	$8(e_{2,0}), 4(e_{0,0}), 2(e_{3,0}), -3(e_{1,0})$	4, 2, 5	$\{e_{2,0}\}; \{e_{2,0}, e_{0,0}\}; \{e_{2,0}, e_{0,0}, e_{3,0}\}$
1	$9(e_{2,1}), 7(e_{3,1}), 5(e_{1,1}), 2(e_{0,1})$	2, 2, 3	$\{e_{2,1}\}; \{e_{2,1}, e_{3,1}\}; \{e_{2,1}, e_{3,1}, e_{1,1}\}$
2	$3(e_{0,2}), 2(e_{2,2}), 2(e_{3,2}), 1(e_{1,2})$	1, 0, 1	$\{e_{0,2}\}; \{e_{0,2}, e_{2,2}\}; \{e_{0,2}, e_{2,2}, e_{3,2}\}$

Total of $3 \times 3 = 9 = nk - k$ pairs (3 per column).

Step 3: Flatten and re-sort by gap size.

Rank r	Gap δ_r	MSV V_r	Origin
0	5	$\{e_{2,0}, e_{0,0}, e_{3,0}\}$	col 0, gap 2
1	4	$\{e_{2,0}\}$	col 0, gap 0
2	3	$\{e_{2,1}, e_{3,1}, e_{1,1}\}$	col 1, gap 2
3	2	$\{e_{2,0}, e_{0,0}\}$	col 0, gap 1
4	2	$\{e_{2,1}\}$	col 1, gap 0
5	2	$\{e_{2,1}, e_{3,1}\}$	col 1, gap 1
6	1	$\{e_{0,2}\}$	col 2, gap 0
7	1	$\{e_{0,2}, e_{2,2}, e_{3,2}\}$	col 2, gap 2
8	0	$\{e_{0,2}, e_{2,2}\}$	col 2, gap 1

Step 4: Barycentric subdivision. Gap sequence: 5, 4, 3, 2, 2, 2, 1, 1, 0.

$$\alpha_r = (r+1)(\delta_r - \delta_{r+1}) : \quad 1, 2, 3, 0, 0, 6, 0, 8, 0.$$

Non-zero terms only:

$$\alpha_0 = 1 \text{ (uses } L_0), \quad \alpha_1 = 2 \text{ (uses } L_1), \quad \alpha_2 = 3 \text{ (uses } L_2), \quad \alpha_5 = 6 \text{ (uses } L_5), \quad \alpha_7 = 8 \text{ (uses } L_7).$$

The Eji_LinComb count matrices $L_r = V_0 + \dots + V_r$ for the non-zero terms are (showing index / count matrix, before canonicalisation):

Term	Count matrix (rows = coordinates $i=0, 1, 2$; cols = vectors $j=0, 1, 2, 3$)
L_0 (idx 1)	$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$
L_1 (idx 2)	$\begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$
L_2 (idx 3)	$\begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$
L_5 (idx 6)	$\begin{pmatrix} 2 & 0 & 3 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$
L_7 (idx 8)	$\begin{pmatrix} 2 & 0 & 3 & 1 \\ 0 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 \end{pmatrix}$

Step 6: Output. $N_{\text{hash}} = 2 \times 3 \times 3 + 1 = 19$.

$$f_2(X) = \left[\underbrace{-3, 2, 1}_{3 \text{ col. min.}}, 1p_0 + 2p_1 + 3p_2 + 6p_5 + 8p_7 \right] \in \mathbb{R}^{22},$$

where each $p_r \in [0, 1]$ ¹⁹.

5 Comparison

Side-by-Side Summary

	Algorithm 1	Algorithm 2
Sort strategy	One global sort of all nk entries	k independent column sorts
Number of gaps	$nk - 1$	$k(n - 1) = nk - k$
MSV structure	Prefix sets of the global order; MSVs can mix entries from any column	Prefix sets within each column; all entries in a given MSV share the same i -index
Hash embedding dim.	$2(nk - 1) + 1 = 2nk - 1$	$2k(n - 1) + 1 = 2nk - 2k + 1$
Anchor values	Global max and global min (2 values)	Per-column minima (k values)
Total output size	$(2nk - 1) + 2 = 2nk + 1$	$(2nk - 2k + 1) + k = 2nk + 1 - k$
Saving vs. Alg 1	—	k dimensions

Why Algorithm 2 is Smaller

Algorithm 2 saves k output dimensions. The saving arises from two interacting changes.

1. **Fewer gaps to embed ($k-1$ fewer).** Sorting per-column gives $k(n - 1)$ gaps; sorting globally gives $nk - 1$ gaps. The difference is $k-1$. Each gap costs 2 hash dimensions (since $N_{\text{hash}} = 2m + 1$), so the hash embedding shrinks by $2(k - 1)$.
2. **More anchor values ($k-2$ more for $k > 2$).** Algorithm 1 stores 2 anchors (global max and min). Algorithm 2 stores k anchors (per-column minima), which is $k - 2$ more for $k > 2$. This partially offsets the hash savings.

Net saving: $2(k - 1) - (k - 2) = k$. This holds for all $k \geq 1$.

The *reason* Algorithm 2 needs $k-1$ fewer gaps is conceptual: in the global sort (Algorithm 1), the very last (smallest) entry has no gap below it within the chain; that “missing” gap is represented by the global minimum anchor. In Algorithm 2, the same thing happens *independently in each column*: the smallest entry of each column has no gap below it in that column, and each such “missing” gap is represented by one of the k column-minimum anchors. Algorithm 2 exploits the per-column structure to reduce the number of “missing” gaps from one (global) to k (one per column), which at first seems *worse* — but the per-column sorting also prevents $k-1$ cross-column comparisons from appearing in the gap list, yielding a net reduction of $k-1$ gaps and ultimately a saving of k output dimensions.

Neither Algorithm is Optimal

Both algorithms embed into $O(nk)$ dimensions, which matches the information content of the input (a multiset of nk real values). However, the exact constants differ; ideally one would embed into just nk reals, and both algorithms use roughly twice that. The factor-of-two overhead reflects the generic-position hashing scheme: each gap is embedded into a two-dimensional “direction” plus a weight, rather than a single real. Reducing this overhead is an open problem.